

Mean Field Approximations

In this PDF the mean field approximations for the Ising and Potts models used in the spininfer library are defined and derived. First the approximations are discussed for the Ising model and subsequently generalized to the Potts model.

1 Ising Model

This section provides a derivation of the mean field approaches for the Ising model. The relevant governing equations are:

$$H(\boldsymbol{\sigma}) = - \sum_i h_i \sigma_i - \sum_{(i,j)} J_{ij} \sigma_i \sigma_j, \quad (1)$$

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} e^{-H(\boldsymbol{\sigma})} = \frac{1}{Z} \exp \left(\sum_i h_i \sigma_i + \sum_i \sum_{j < i} J_{ij} \sigma_i \sigma_j \right), \quad (2)$$

$$Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}, \quad (3)$$

where Z is the partition function and $\boldsymbol{\sigma} \in \{-1, 1\}^N$ represents a system of N spins.

1.1 Mean field setup

Now consider that the spin of site i can be represented by its mean and the fluctuations:

$$\sigma_i = \langle \sigma_i \rangle + \delta \sigma_i, \text{ where } \langle \delta \sigma_i \rangle = 0. \quad (4)$$

Using this we can write the correlations as follows:

$$\sigma_i \sigma_j = \langle \sigma_i \rangle \langle \sigma_j \rangle + \langle \sigma_i \rangle \delta \sigma_j + \langle \sigma_j \rangle \delta \sigma_i + \delta \sigma_i \delta \sigma_j, \quad (5)$$

which can be rewritten to

$$\sigma_i \sigma_j = \langle \sigma_i \rangle \sigma_j + \langle \sigma_j \rangle \sigma_i - \langle \sigma_i \rangle \langle \sigma_j \rangle + \delta \sigma_i \delta \sigma_j. \quad (6)$$

1.2 Naive mean field approximation

In the naive mean field approximation we assume that the product of the fluctuations is small $\delta \sigma_i \delta \sigma_j \approx 0$. Using this, we can substitute this approximation into the Hamiltonian noting that J is symmetric, which gets rid of the spin-spin dependence:

$$H_{\text{MF}}(\boldsymbol{\sigma}) = - \sum_i h_i \sigma_i - \sum_{(i,j)} J_{ij} \langle \sigma_i \rangle \sigma_j - \sum_{(i,j)} J_{ij} \langle \sigma_j \rangle \sigma_i + \sum_{(i,j)} J_{ij} \langle \sigma_i \rangle \langle \sigma_j \rangle \quad (7)$$

$$= - \sum_i \left(h_i + \sum_{j \neq i} J_{ij} \langle \sigma_j \rangle \right) \sigma_i + \sum_{(i,j)} J_{ij} \langle \sigma_i \rangle \langle \sigma_j \rangle \quad (8)$$

$$= - \sum_i h_i^{\text{eff}} \sigma_i + \sum_{(i,j)} J_{ij} \langle \sigma_i \rangle \langle \sigma_j \rangle, \quad (9)$$

where $h_i + \sum_{j \neq i} J_{ij} \langle \sigma_j \rangle = h_i^{\text{eff}}$. Using this the partition function can be rewritten to:

$$Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})} \approx e^{-\sum_{(i,j)} \langle \sigma_i \rangle \langle \sigma_j \rangle J_{ij}} \sum_{\boldsymbol{\sigma}} e^{\sum_i h_i^{\text{eff}} \sigma_i} \quad (10)$$

$$= e^{-\sum_{(i,j)} \langle \sigma_i \rangle \langle \sigma_j \rangle J_{ij}} \sum_{\boldsymbol{\sigma}} \prod_i e^{h_i^{\text{eff}} \sigma_i} \quad (11)$$

$$= e^{-\sum_{(i,j)} \langle \sigma_i \rangle \langle \sigma_j \rangle J_{ij}} \prod_i \sum_{\sigma_i \in \{1, -1\}} e^{h_i^{\text{eff}} \sigma_i} \quad (12)$$

$$= 2^N e^{-\sum_{(i,j)} \langle \sigma_i \rangle \langle \sigma_j \rangle J_{ij}} \prod_i \cosh h_i^{\text{eff}}. \quad (13)$$

Now, to get the mean of a single site i we use the identity:

$$\frac{\partial \log(Z)}{\partial h_i} = \langle \sigma_i \rangle, \quad (14)$$

Because the spins decouple under H_{MF} , the thermal expectation value of spin σ_i is evaluated directly from its single-site distribution:

$$\langle \sigma_i \rangle = \sum_{\sigma_i \in \{-1, 1\}} \sigma_i p_i(\sigma_i) = \frac{e^{h_i^{\text{eff}}} - e^{-h_i^{\text{eff}}}}{e^{h_i^{\text{eff}}} + e^{-h_i^{\text{eff}}}} = \tanh(h_i^{\text{eff}}). \quad (15)$$

These are the self-consistent mean-field equations:

$$\langle \sigma_i \rangle = \tanh \left(h_i + \sum_{j \neq i} J_{ij} \langle \sigma_j \rangle \right). \quad (16)$$

Now, to get the inverse parameters, we first have to define the following identity

$$\frac{\partial \langle \sigma_i \rangle}{\partial h_j} = \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_i \sigma_j e^{-H(\boldsymbol{\sigma})} - \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_i e^{-H(\boldsymbol{\sigma})} \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_j e^{-H(\boldsymbol{\sigma})} \quad (17)$$

$$= \langle \sigma_i \sigma_j \rangle - \langle \sigma_i \rangle \langle \sigma_j \rangle = C_{ij}. \quad (18)$$

Using this we can take the derivative of the self-consistent equations towards h_j :

$$\frac{\partial \langle \sigma_i \rangle}{\partial h_j} = (1 - \langle \sigma_i \rangle^2) (\delta_{ij} + \sum_{i \neq k} J_{ik} C_{kj}). \quad (19)$$

This can be rewritten to

$$\frac{C_{ij}}{1 - \langle \sigma_i \rangle^2} = \delta_{ij} + \sum_{i \neq k} J_{ik} C_{kj}, \quad (20)$$

which using D being the diagonal matrix defined by $\frac{1}{1 - \langle \sigma_i \rangle^2}$ in matrix notation simplifies to:

$$DC = I + JC, \quad (21)$$

where J is isolated accordingly

$$J = D - C^{-1}. \quad (22)$$

Now that J is determined the parameter h follows trivially from the self-consistent equations.

2 Potts Model

For a system of N variables where each spin takes values in $\sigma_i \in \{1, \dots, q\}$, let the indicator variable for site i adopting state a be defined as:

$$\sigma_i(a) \equiv \delta(\sigma_i, a) = \begin{cases} 1 & \text{if } \sigma_i = a, \\ 0 & \text{otherwise.} \end{cases} \quad (23)$$

The generalized Potts Hamiltonian can be written in terms of these indicator variables as:

$$H(\boldsymbol{\sigma}) = - \sum_{i=1}^N \sum_{a=1}^q h_i(a) \sigma_i(a) - \sum_{(i,j)} \sum_{a=1}^q \sum_{b=1}^q J_{ij}(a,b) \sigma_i(a) \sigma_j(b), \quad (24)$$

where $h_i(a)$ is the local field for state a at site i , and $J_{ij}(a,b)$ is the pairwise coupling between site i in state a and site j in state b , satisfying $J_{ij}(a,b) = J_{ji}(b,a)$. Note the change in notation from the rest of the derivations, which makes the mean field approximation easier to follow.

The probability of a configuration $\boldsymbol{\sigma}$ is given by the Boltzmann distribution:

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} e^{-H(\boldsymbol{\sigma})}, \text{ where } Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}. \quad (25)$$

2.1 Naive Mean Field Approximation

In the naive mean-field approximation, we assume that second-order fluctuations are negligible, $\delta\sigma_i(a)\delta\sigma_j(b) \approx 0$. Substituting this approximation into the Potts Hamiltonian gives:

$$H(\boldsymbol{\sigma}) \approx - \sum_{i=1}^N \sum_{a=1}^q h_i(a) \sigma_i(a) \quad (26)$$

$$- \sum_{(i,j)} \sum_{a=1}^q \sum_{b=1}^q J_{ij}(a,b) \left(\sigma_i(a) \langle \sigma_j(b) \rangle + \sigma_j(b) \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_j(b) \rangle \right), \quad (27)$$

which decouples the spins as follows:

$$H_{\text{MF}}(\boldsymbol{\sigma}) = - \sum_{i=1}^N \sum_{a=1}^q h_i^{\text{eff}}(a) \sigma_i(a) + \sum_{(i,j)} \sum_{a=1}^q \sum_{b=1}^q J_{ij}(a,b) \langle \sigma_i(a) \rangle \langle \sigma_j(b) \rangle, \quad (28)$$

where the effective local field $h_i^{\text{eff}}(a)$ is given by:

$$h_i^{\text{eff}}(a) = h_i(a) + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(a,b) \langle \sigma_j(b) \rangle. \quad (29)$$

Using the mean field Hamiltonian and rewriting the constant term as E_0 , we can write the mean field partition function as:

$$Z_{\text{MF}} = \sum_{\boldsymbol{\sigma}} e^{-H_{\text{MF}}(\boldsymbol{\sigma})} \quad (30)$$

$$= e^{-E_0} \sum_{\boldsymbol{\sigma}} \exp \left(\sum_{i=1}^N \sum_{a=1}^q h_i^{\text{eff}}(a) \sigma_i(a) \right) \quad (31)$$

$$= e^{-E_0} \sum_{\boldsymbol{\sigma}} \prod_{i=1}^N \prod_{a=1}^q e^{h_i^{\text{eff}}(a) \sigma_i(a)} \quad (32)$$

$$= e^{-E_0} \prod_{i=1}^N \left(\sum_{\sigma_i=1}^q \prod_{a=1}^q e^{h_i^{\text{eff}}(a) \delta_{\sigma_i, a}} \right) \quad (33)$$

$$= e^{-E_0} \prod_{i=1}^N \left(\sum_{a=1}^q e^{h_i^{\text{eff}}(a)} \right), \quad (34)$$

which yields the partition function that can be decomposed into its single site partition functions. Because the spins decouple under H_{MF} , the expectation value of the indicator variable $\sigma_i(a)$ is given directly by its distribution:

$$\langle \sigma_i(a) \rangle = p_i(\sigma_i = a) = \frac{e^{h_i^{\text{eff}}(a)}}{\sum_{b=1}^q e^{h_i^{\text{eff}}(b)}} = \frac{\exp \left(h_i(a) + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(a, b) \langle \sigma_j(b) \rangle \right)}{\sum_{c=1}^q \exp \left(h_i(c) + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(c, b) \langle \sigma_j(b) \rangle \right)}, \quad (35)$$

which defines the self-consistent mean-field equations for the Potts model.

Similar to the Ising model section, we will derive the following identity:

$$\frac{\partial \langle \sigma_i(a) \rangle}{\partial h_j(b)} = \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_i(a) \sigma_j(b) e^{-H(\boldsymbol{\sigma})} - \frac{1}{Z^2} \sum_{\boldsymbol{\sigma}} \sigma_i(a) e^{-H(\boldsymbol{\sigma})} \sum_{\boldsymbol{\sigma}'} \sigma_j(b) e^{-H(\boldsymbol{\sigma}')} \quad (36)$$

$$= \langle \sigma_i(a) \sigma_j(b) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_j(b) \rangle = C_{ij}(a, b), \quad (37)$$

which is the 4-dimensional correlation matrix. Taking this same derivative with respect to the self-consistent equations, we get:

$$\frac{\partial \langle \sigma_i(a) \rangle}{\partial h_k(d)} = \sum_{c=1}^q \frac{\partial \langle \sigma_i(a) \rangle}{\partial h_i^{\text{eff}}(c)} \frac{\partial h_i^{\text{eff}}(c)}{\partial h_k(d)}. \quad (38)$$

Now, first we evaluate

$$\frac{\partial \langle \sigma_i(a) \rangle}{\partial h_i^{\text{eff}}(c)} = \delta_{ac} \frac{e^{h_i^{\text{eff}}(a)}}{\sum_{b=1}^q e^{h_i^{\text{eff}}(b)}} - \frac{e^{h_i^{\text{eff}}(a)} e^{h_i^{\text{eff}}(c)}}{\left(\sum_{b=1}^q e^{h_i^{\text{eff}}(b)} \right)^2} = \delta_{ac} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(c) \rangle. \quad (39)$$

Now, for the derivative of the effective field, we get:

$$\frac{\partial h_i^{\text{eff}}(c)}{\partial h_k(d)} = \delta_{ik} \delta_{cd} + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(c, b) \frac{\partial \langle \sigma_j(b) \rangle}{\partial h_k(d)} = \delta_{ik} \delta_{cd} + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(c, b) C_{jk}(b, d). \quad (40)$$

In total this yields:

$$C_{ik}(a, d) = \sum_{c=1}^q (\delta_{ac} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(c) \rangle) (\delta_{ik} \delta_{cd} + \sum_{j \neq i} \sum_{b=1}^q J_{ij}(c, b) C_{jk}(b, d)). \quad (41)$$

Now, expanding the sum over c , the first term evaluates to:

$$\sum_{c=1}^q (\delta_{ac} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(c) \rangle) \delta_{ik} \delta_{cd} = \delta_{ik} (\delta_{ad} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(d) \rangle), \quad (42)$$

and for the second term, we get:

$$\sum_{c=1}^q (\delta_{ac} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(c) \rangle) \sum_{j \neq i} \sum_{b=1}^q J_{ij}(c, b) C_{jk}(b, d) \quad (43)$$

$$= \langle \sigma_i(a) \rangle \sum_{j \neq i} \sum_{b=1}^q \left(J_{ij}(a, b) - \sum_{c=1}^q \langle \sigma_i(c) \rangle J_{ij}(c, b) \right) C_{jk}(b, d). \quad (44)$$

In total this yields:

$$C_{ik}(a, d) = \delta_{ik} (\delta_{ad} \langle \sigma_i(a) \rangle - \langle \sigma_i(a) \rangle \langle \sigma_i(d) \rangle) + \langle \sigma_i(a) \rangle \sum_{j \neq i} \sum_{b=1}^q \left(J_{ij}(a, b) - \sum_{c=1}^q \langle \sigma_i(c) \rangle J_{ij}(c, b) \right) C_{jk}(b, d), \quad (45)$$

where the first term is the block-diagonal single-site covariance matrix $(D^{-1})_{ik}(a, d) = \delta_{ik} C_{ii}(a, d)$. Setting $J_{ii} = 0$, this can be written in block matrix notation as:

$$C = D^{-1} + D^{-1}JC, \tag{46}$$

which isolating J gives

$$J = D - C^{-1}. \tag{47}$$

The $h_i(a)$ parameters can now again be trivially computed from the self-consistent equations.

2.2 Gauge freedom

Since the Potts model is overparametrized, the inversion of C will not work. That is why in the library the last state is dropped as a reference state, the parameters are determined and then are added back as zeros. The gauge used then determines these parameters.